

MID TERM EXAMINATION
B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)
 Third Semester, September - October, 2025

Paper Code: CIC-211**Subject: Object-Oriented Programming Using C++****Time: 1 ½ Hrs.****Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1	Attempt the following:	2	CO1
	(a) Compare and contrast the structured programming and object-oriented programming.	2	CO1
	(b) Demonstrate any 2 uses of scope resolution operator with appropriate examples.	2	CO2
	(c) How does C++ support data abstraction and Encapsulation? Appraise with an appropriate example why is it necessary to create good abstraction?	2	CO2
	(d) Write short note on dynamic memory allocation in C++.	2	CO2
2	(e) Explain the state, identity and behaviour of an object.		
3	(a) Using the concept of function overloading write function for calculating the area of triangle, circle and rectangle.	5	CO1
	(b) Demonstrate parameterized constructor and copy constructor with an example. Why an argument to a copy constructor should be passed by reference?	5	CO2
4	(a) List the advantages and disadvantages of inline functions. How are inline member function defined outside the body of a class?	5	CO1
	(b) Explain briefly the concept of operator overloading. Write a code in C++ to overload addition operator to add two complex numbers.	5	CO2
5	(a) Explain friend function and its functionality using suitable example. Why friend functions should be avoided?	5	CO1
	(b) Explain the concept of static member functions in C++ with suitable examples. How do they differ from normal member functions?	5	CO2